

Eric Forbes

5G Systems Integration Lead · Lockheed Martin · PE, PMP, PhD Candidate
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PROFILE

Systems integration and wireless product leader grown from bench-level test engineering to directing multi-vendor O-RAN deployments, sub-contractor teams, and on-site customer integrations for mission-critical systems. I own delivery from architecture through integration, test, and validation, working top down with hands-on depth at every layer beneath. I build teams that make products, investing in long-term strategy and team culture for full system lifecycles. I work across wireless link types to deploy and leverage the right technology for the CONOPS at hand, applying systems engineering methodology to ensure requirements, scope, and business constraints are met.

EXPERIENCE

Lockheed Martin

5G Systems I&T Lead, RMS

May 2021 – Present

November 2024 – Present

Product owner for O-RAN and Tactical MANET deployments connecting mission-critical applications.

- Set technical direction for O-RAN software deployment on low-SWaP platforms; direct and coordinate sub-contractor teams across a multi-vendor environment.
- Own system architecture models through to hands-on, on-site integration and validation for internal and external customers.

5G Systems I&T Engineer, RMS

April 2024 – November 2024

Demonstrated the world's first 5G VPX CMOSS system with open-source software.

- Executed L2/L3 network integration, PTP timing analysis, 5G RAN/Core log debugging, and end-to-end system testing.

4G Systems Engineer, Space

August 2022 – April 2024

Brought the first LTE GEO Satellite system to close.

- Performed full-stack eNB analysis (UE modem & eNB); developed test plans and authored program documentation.
- Applied Python for large dataset analysis; Bash scripting for system debugging and automation.

5G Systems Engineer, Space

May 2021 – August 2022

Defined architecture and requirements for 5G Non-Terrestrial Networks (NTN).

- Developed NTN requirements and network diagrams; tracked 3GPP Rel-17 NTN working groups to position the product for where the standard was heading.

L3Harris Technologies

February 2020 – May 2021

Sr. Specialist, Systems Engineer

Requirements Lead for a new Radar Warning Receiver (RWR) System program.

- Owned requirements development and decomposition in IBM DOORS; engaged prime customers for requirements review and program analysis.
- Applied MagicDraw & SysML for MBSE across the Space and Airborne Systems division.

G3 Technologies, Inc.

January 2017 – February 2020

Cellular Systems Test Engineer

Protocol testing across four generations of wireless standards for commercial and defense customers.

- Debugged and tested per 3GPP/3GPP2 standards (LTE, UMTS, GSM, CDMA); analyzed Qualcomm chipsets (QXDM).
- Performed physical layer (RF) testing and verification; authored full system manuals and customer documentation.

Intertek

January 2015 – January 2017

Project Engineer

Led compliance testing and certification for consumer electronics to ANSI, CA, and IEC safety standards.

- Managed customer timelines, executed test plans, and delivered compliance reports for end-product certification.
- Served as Safety Officer; completed OSHA 40-hour training.

EDUCATION

Stevens Institute of Technology

2020 – Fall 2027 (Expected)

Doctor of Philosophy (PhD), Systems Engineering: 5G

Stevens Institute of Technology

2020 – 2024

Graduate Certificate, Space Systems Engineering

New Jersey Institute of Technology

2014 – 2016

Master of Science (MS), Electrical Engineering: Wireless Communications

Penn State University

2010 – 2014

Bachelor of Science (BS), Electrical Engineering

CERTIFICATIONS

Project Management Professional (PMP), May 2022 · Licensed Professional Engineer (PE), New York, Dec. 2019: Electronics, Controls & Communications · Fundamentals of Engineering (FE), Jan. 2017 · IEEE Wireless Communications Professional (WCP), Oct. 2018

PUBLICATIONS

- [1] E. Forbes *et al.*, "Coexistent Evaluation on the Effects of Radar PRF and 5G Symbol-Based Scheduling," *IEEE MILCOM*, Los Angeles, CA, Oct. 2025. **Poster Session Honorable Mention.**
- [2] E. Forbes *et al.*, "5G and Radar Coexistence: Demonstrating Pulsed Radar Interference and 5G Performance Metrics," *IEEE MILCOM Demo Session*, Oct. 2025.
- [3] E. Forbes *et al.*, "Cellular Data-Based Indoor Localization for Smart Health Applications," *IEEE CHASE*, Jun. 2025.
- [4] E. Forbes *et al.*, "Communication Systems for Autonomous eVTOL: Regulatory Challenges, Emerging Technologies, and Future Directions," *IEEE*, 2025.
- [5] E. Forbes, E. Silverstein, and Y. Wang, "Elevation-Aware BLER-Centric Link Adaptation for Dynamic LEO 5G NTN Systems," *2026 IEEE Aerospace Conference*, 2026.